NON-INVASIVE ANAEMIA DETECTOR

The Mini Project Report submitted in partial fulfilment of the requirements for the award of the Degree of

Bachelor of Technology

in

Electronics and Communication Engineering

under the

APJ Abdul Kalam Technological University

by

SWETHA ANIL – MGP23EC066

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SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS)

DEPARTMENT OF ELECTRONICS ENGINEERING

KOTTAYAM, KERALA (INDIA)

April 2026

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DEPARTMENT OF ELECTRONICS ENGINEERING

KOTTAYAM, KERALA (INDIA)

BONAFIDE CERTIFICATE

This is to certify that the project entitled **Non-Invasive Anaemia Detector** submitted by **Swetha Anil (MGP23EC066),** for the award of the Bachelor of Technology in Electronics and Communication Engineering is a Bonafide record of the Mini Project carried out by her under my guidance and supervision at Saintgits College of Engineering (Autonomous). This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

I, Swetha Anil (MGP23EC066), hereby declare that this project report entitled "Non Invasive Anaemia Detector" is the record of the original work done by me under the guidance of Er. Jyothish Chandran G, Assistant Professor, Department of Electronics Engineering, Saintgits College of Engineering (Autonomous). To the best of my knowledge, this work has not formed the basis for the award of any degree, diploma, fellowship, or a similar award to any candidate in any University.

Place: Kottayam
Date: April 2026

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I would like to express my sincere gratitude to Almighty God for His blessings, guidance, and strength, which helped me successfully complete this project.

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Abbreviations

Hb	Haemoglobin
PPG	Photoplethysmogram
AC	Alternating Current Component (Pulsatile Signal)
DC	Direct Current Component (Baseline Signal)
IR	Infrared
LED	Light Emitting Diode
OLED	Organic Light Emitting Diode
MAX30102	Optical Pulse Oximeter and Heart-Rate Sensor Module
ESP32	Espressif Systems 32-bit Microcontroller
I2C	Inter-Integrated Circuit
SDA	Serial Data Line
SCL	Serial Clock Line
IDE	Integrated DevelopmentEnvironment
API	Application Programming Interface
g/dL	Grams per Decilitre nm
R	Ratio of Red and Infrared Signal Components

Abstract

Anaemia is a widespread health condition characterized by low haemoglobin (Hb) concentration and is traditionally diagnosed through invasive blood tests. This project focuses on developing a low-cost, non-invasive anaemia detection system based on optical sensing principles, suitable for point-of-care applications. The system operates using the Beer– Lambert Law, employing red (660 nm) and infrared (940 nm) wavelengths to study haemoglobin absorption characteristics through a fingertip.

A MAX30102 optical sensor is used to capture Photoplethysmogram (PPG) signals. From these signals, the AC and DC components of both red and infrared light are extracted. Basic digital signal processing techniques, including filtering, DC removal, and peak detection, are applied to minimize noise and motion artifacts, ensuring more accurate signal interpretation. A regression-based calibration model is developed by correlating the extracted optical features with reference haemoglobin values obtained from standard blood tests. The system estimates haemoglobin levels in g/dL and displays them along with visual indicators for normal and anaemic conditions. The prototype demonstrates the feasibility of non-invasive haemoglobin estimation using affordable hardware, making it a useful proof-of-concept for biomedical instrumentation, though not intended for clinical diagnosis.

Keywords: Anaemia Detection, Non-Invasive Measurement, MAX30102, ESP32, Optical Sensing, Haemoglobin Estimation.

Chapter 1

Introduction

1.1 Background

Anaemia is a common health condition characterized by a low concentration of haemoglobin (Hb) in the blood, reducing the oxygen-carrying capacity of red blood cells. It affects a large portion of the global population, especially in developing countries, and can lead to fatigue, weakness, and serious health complications if left undetected. Conventional diagnosis of anaemia relies on invasive blood tests, which can be uncomfortable, time-consuming, and not easily accessible in remote or resource-limited areas.

With advancements in biomedical instrumentation, there is growing interest in developing non-invasive and cost-effective methods for health monitoring. Optical sensing techniques provide a promising solution, allowing the estimation of physiological parameters without the need for blood extraction. This project leverages such techniques to create a simple and portable anaemia detection system.

1.2 Motivation

Traditional methods for anaemia detection require laboratory-based blood analysis, which may not be practical for frequent or large-scale screening. This creates several limitations:

- **Invasive procedure**: Blood collection causes discomfort and may discourage regular monitoring.
- **Limited accessibility**: Rural and low-resource areas may lack proper medical facilities.
- **Time-consuming process**: Laboratory analysis requires time and trained personnel.
- **Lack of portability**: Conventional devices are not suitable for point-of-care or home use.

These challenges highlight the need for a non-invasive, affordable, and portable system that can provide quick haemoglobin estimation. Such a system would enable early detection and continuous monitoring, improving healthcare accessibility.

1.3 Project Overview

This project presents a non-invasive anaemia detection system based on optical sensing principles using the Beer–Lambert Law. The system uses two wavelengths—red (660 nm) and infrared (940 nm)—to analyse haemoglobin absorption characteristics through a fingertip.

A **MAX30102** optical sensor is used to acquire Photoplethysmogram (PPG) signals. From these signals, the AC and DC components of both red and infrared light are extracted. Basic digital signal processing techniques such as filtering, DC removal, and peak detection are applied to reduce noise and motion artifacts.

A regression-based calibration model is developed by correlating the extracted optical features with reference haemoglobin values obtained from standard blood tests. The system estimates haemoglobin levels in g/dL and displays them along with indicators for normal and anaemic conditions, demonstrating a practical proof-of-concept for non-invasive screening.

1.4 Scope of the Project

The scope of this project includes:

- Design and development of a non-invasive haemoglobin estimation system using optical sensing.
- Implementation of signal acquisition using the MAX30102 sensor.
- Application of basic digital signal processing techniques for noise reduction.
- Development of a regression model for haemoglobin estimation.
- Display of results with clear indicators for anaemia detection.

The system is intended as a prototype for educational and research purposes, showcasing the potential of optical sensing in biomedical applications

1.5 Organisation of the Report

The report is organized as follows: Chapter 1 introduces the background, motivation, and overview of the project. Chapter 2 discusses the theoretical concepts, including the Beer–

Lambert Law and optical sensing principles. Chapter 3 explains the system design and hardware components used. Chapter 4 covers signal processing techniques and model development. Chapter 5 presents results and analysis. Finally, Chapter 6 concludes the work and suggests future improvements.

Chapter 2

Literature Review

2.1 Introduction

The combination of biomedical engineering, optical sensing, and embedded systems has led to the development of non-invasive health monitoring techniques. In recent years, researchers have focused on replacing traditional invasive blood tests with optical methods for estimating physiological parameters such as haemoglobin concentration. This chapter reviews existing literature on non-invasive anaemia detection using optical sensors and signal processing techniques. It highlights the strengths and limitations of current approaches and identifies gaps that motivate the proposed system.

2.2 Optical-Based Haemoglobin Monitoring

Several researchers have explored the use of optical principles, particularly the Beer–Lambert Law, for estimating haemoglobin levels. These systems typically use red and infrared light to measure absorption characteristics of blood through body tissues such as the fingertip or earlobe. Studies show that haemoglobin concentration affects light absorption, making it possible to estimate Hb levels non-invasively. Many systems use photoplethysmography (PPG) signals to extract useful features

. *Limitation*: Accuracy can be affected by skin thickness, motion artifacts, and ambient light interference, making calibration necessary.

2.3 PPG Sensor-Based Detection Systems

Devices using PPG sensors like MAX30102 have been widely used for measuring heart rate and oxygen saturation, and recent research extends their application to haemoglobin estimation. These systems extract AC and DC components of red and infrared signals and use them to compute ratios related to blood properties.

Microcontrollers are used to process signals and apply filtering techniques to reduce noise. Some systems also integrate machine learning or regression models for better estimation.

Observation: PPG-based systems are low-cost and portable, but require proper signal processing to ensure reliable results.

2.4 Non-Invasive Health Monitoring Systems

Recent developments include portable and wearable devices that monitor multiple health parameters non-invasively. These systems often provide real-time results and are designed for point-of-care or home-based monitoring. Some solutions integrate mobile applications for data display and tracking, improving accessibility and user convenience.

Limitation: Many systems depend on complex hardware or smartphone connectivity, increasing cost and design complexity.

2.5 Signal Processing and Calibration Techniques

Accurate haemoglobin estimation depends heavily on signal processing and calibration methods. Techniques such as filtering, DC removal, and peak detection are used to clean PPG signals and extract meaningful features.

Regression models are commonly used to correlate optical measurements with actual haemoglobin values obtained from blood tests. Calibration improves accuracy but requires reference data.

Key Insight: Combining multiple signal features improves estimation accuracy compared to using a single parameter.

2.6 Summary and Research Gap

From the literature review, it is clear that non-invasive haemoglobin estimation is feasible using optical sensing techniques. However, many existing systems face challenges such as high cost, dependency on complex algorithms, or lack of portability. There is a need for a simple, low-cost, and portable system that provides reasonably accurate haemoglobin estimation without requiring complex infrastructure. The proposed project addresses this gap by using a MAX30102 sensor, basic signal processing, and a regression-based model to develop an affordable and easy-to-use anaemia detection system. The summary of related work are mentioned in the Table 2.1.

Table 2.1: Summary of Related Work

No.	Authors	Technique	Limitation
1	Smith et al. (2020)	Optical sensing using Beer–Lambert Law with red & IR LEDs	Affected by motion artifacts and ambient light noise
2	Patel et al. (2021)	PPG-based haemo-globin estimation using MAX30102 sensor	Requires proper cali-bration for accurate results
3	Lee et al. (2022)	Machine learning model for non-invasive Hb predic-tion	High computational complexity
4	Kumar & Singh (2023)	Smartphoneinte-grated anaemia detec-tion system	Depends on mobile app and internet con-nectivity
5	Rahman et al. (2022)	Multi-parameter health monitoring us-ing optical sensors	Increased system cost and design com-plexity

Chapter 3

Objectives

The primary objective of this project is to design and implement a Non-Invasive Anaemia Detection System that utilizes optical sensing and embedded systems to provide real-time haemoglobin estimation. The system aims to:

- Provide a painless and quick method for anaemia screening.
- Reduce dependence on invasive blood testing methods.
- Improve accessibility to basic health monitoring, especially in remote areas.

By integrating optical sensors, microcontroller-based processing, and calibration techniques, the project aims to develop a reliable and cost-effective solution for haemoglobin estimation. The system is designed to be portable, user-friendly, and efficient, making it suitable for real time health monitoring applications.

3.1 Real-Time Haemoglobin Detection

Develop a system capable of estimating haemoglobin levels in real time using optical sensing technique .Continuously acquire red and infrared light signals from the fingertip and process them to obtain reliable measurements.

3.2 Optical Signal Acquisition

Utilize the MAX30102 sensor to capture red and infrared light absorption data from human tissue. Ensure proper finger detection and stable signal acquisition for accurate readings

3.3 Data Processing and Calibration

Process the acquired signals using averaging and filtering techniques to reduce noise. Apply a calibration equation derived from experimental data to estimate haemoglobin

3.4 Haemoglobin Classification

Classify the estimated haemoglobin values into categories such as normal, mild, moderate, and severe anaemia. Provide meaningful interpretation of results based on standard medical thresholds.

3.5 Display and User Interaction

Develop a user-friendly interface using an OLED display to show haemoglobin levels and status. Implement a push-button mechanism to initiate measurement and ensure ease of use.

3.6 System Accuracy and Reliability

Improve accuracy through calibration and offset correction techniques. Ensure consistent performance under varying conditions such as finger placement and ambient light.

3.7 Portability and Cost Effectiveness

Design a compact and portable system suitable for everyday use. Ensure low-cost implementation for accessibility in resource-limited settings.

3.8 Scalability and Enhancement

Develop a system that can be further enhanced with additional features such as gender based calibration and skin tone analysis. Enable future integration with advanced algorithms for improved accuracy.

3.9 User-Friendly Operation

Ensure the system is simple to operate without requiring technical expertise. Provide clear and real-time output for easy understanding

Chapter 4

Methodology

The methodology begins with understanding the principles of non-invasive haemoglobin estimation using optical sensing techniques, focusing on how red and infrared light interact with blood and how haemoglobin concentration influences light absorption. Based on this, the system is designed to capture real-time optical signals from the fingertip using the MAX30102 sensor, ensuring reliable data acquisition under proper conditions. The collected signals are processed using the ESP32 microcontroller, where filtering and averaging techniques are applied to reduce noise and improve accuracy. The ratio of red to infrared signals is then calculated and used to estimate haemoglobin levels through a calibration equation derived from experimental data. The system further classifies haemoglobin values into different anaemia levels and displays the results in real time on an OLED screen. The design emphasizes portability, cost-effectiveness, and ease of use, making it suitable for preliminary health screening. By following this structured approach, the project provides a reliable and efficient solution for non-invasive anaemia detection.

4.1 System Architecture

4.1.1 Overview

The system architecture comprises three fundamental components: Optical Signal Acquisition, Data Processing and Communication, and Output Display. Each component plays a crucial role in ensuring accurate and reliable haemoglobin estimation. The system utilizes a MAX30102 optical sensor connected to an ESP32 microcontroller, which processes the acquired signals and displays the results on an OLED screen. The architecture ensures real-time measurement, efficient data processing, and user-friendly output for anaemia detection.

4.1.2 Signal Acquisition and Sensor Deployment

The first step involves capturing optical signals from the user's fingertip using the MAX30102 sensor. The sensor emits red and infrared light into the tissue and measures the reflected light intensity. Proper finger placement is ensured to obtain stable and accurate readings. The system continuously collects multiple samples to improve reliability and reduce noise in the measurements.

4.1.3 Data Processing Setup

Once the optical signals are acquired, they are transmitted to the ESP32 microcontroller via I2C communication. The microcontroller processes the data by applying filtering and averaging techniques to remove noise and improve accuracy. The ratio of red to infrared signals is calculated and used for further analysis.

4.1.4 Haemoglobin Estimation and Decision Making

The processed ratio is converted into haemoglobin concentration using a calibration equation derived from experimental data. A decision-making mechanism classifies the haemoglobin level into categories such as normal, mild, moderate, and severe anaemia. The system is designed to be adaptable, allowing future improvements such as advanced algorithms and additional parameters for enhanced accuracy.

4.2 Circuit Development

The block diagram of the system (Figure 4.1) illustrates the signal flow between subsystems**:**

4.2.1 Hardware

The hardware development phase is essential for the proper functioning of the non-invasive anaemia detection system. It involves integrating the sensor, microcontroller, display, and input components.

Sensor module:

- **MAX30102 sensor** – Measures red and infrared light absorption from the fingertip.

Microcontroller:

- **ESP32** – Processes sensor data and controls the system operations.

Display unit:

- **OLED display** – Shows haemoglobin values and anaemia status.

Input control:

- **Push button** – Initiates measurement process.

Power management:

The system is powered using a stable power supply, ensuring efficient operation with minimal energy consumption.

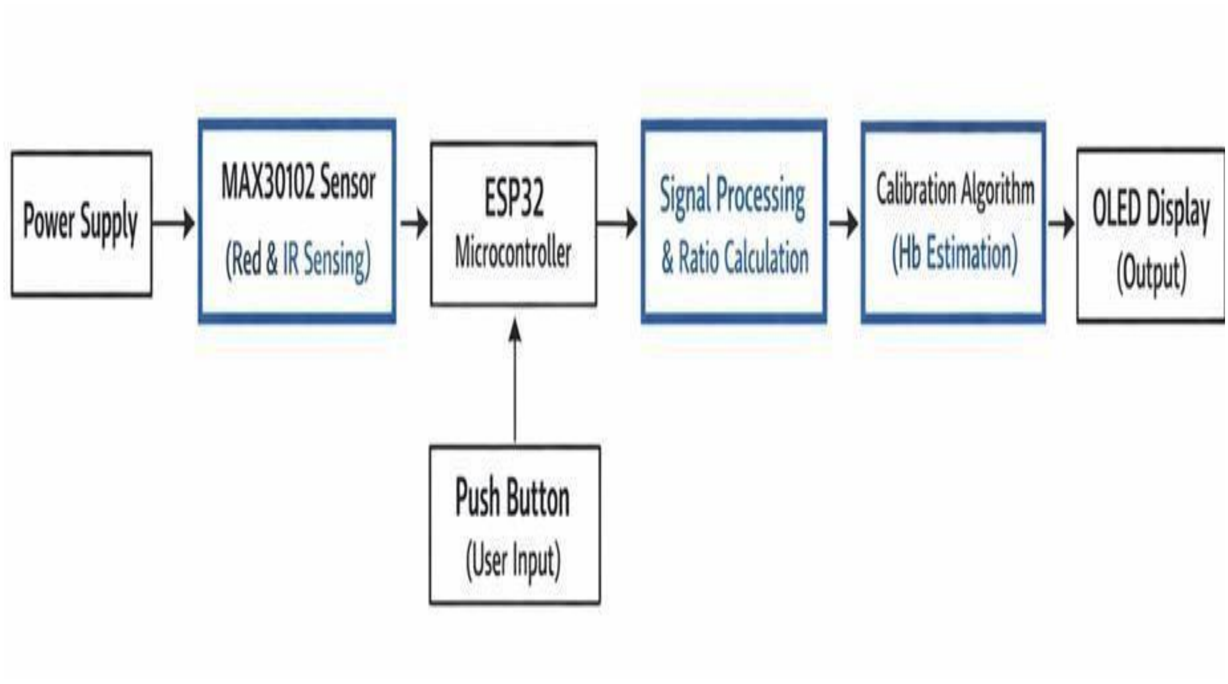


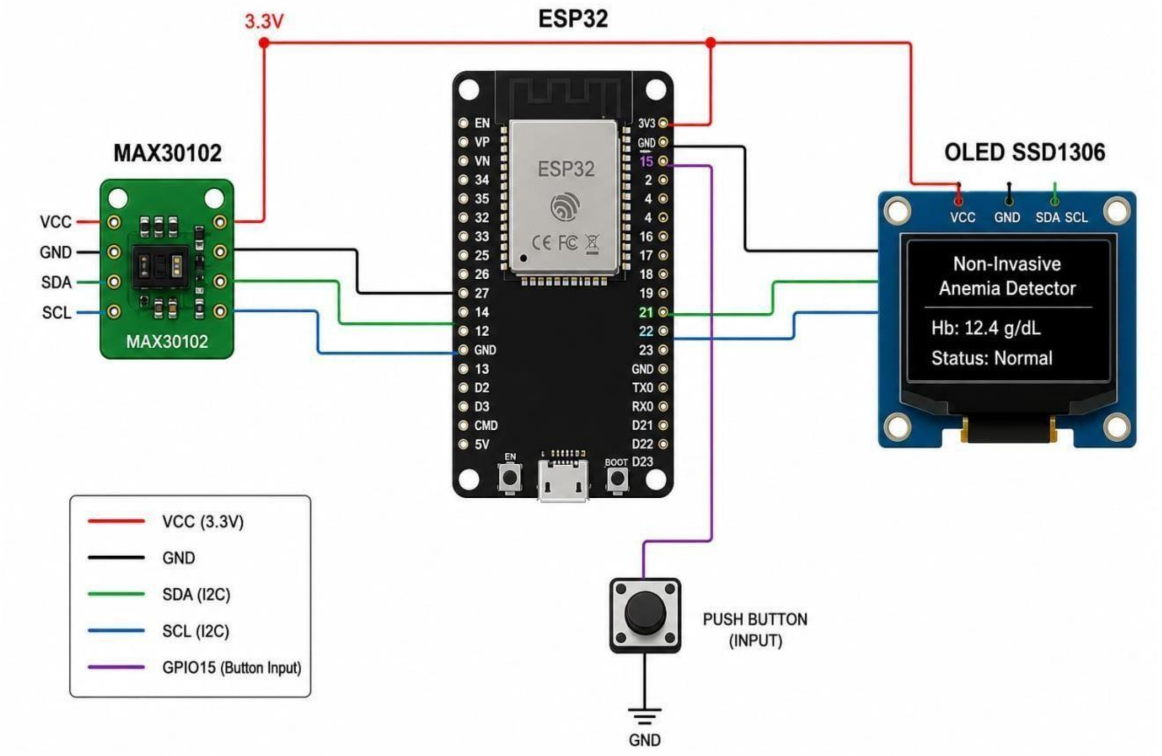
Figure 4.1 System Block Diagram

4.2.2 Circuit Diagram

The system uses a MAX30102 sensor connected to a microcontroller (Arduino/ESP32) via I2C communication (SDA and SCL pins). The sensor operates at 3.3V and shares a common ground with the controller. It emits red and infrared light through the fingertip and detects the reflected signals. The microcontroller processes these signals to estimate haemoglobin levels and displays the result.

The complete circuit configuration of the proposed system, including sensor interfacing and alert components, is illustrated in Figure 4.2.

Figure 4.1 System Block Diagram



Chapter 5

Embedded Firmware and Logic

5.1 Introduction

The firmware is the core software component of the non-invasive anaemia detection system. It processes raw signals obtained from the optical sensor and converts them into meaningful haemoglobin (Hb) values. The firmware also controls data acquisition, signal processing, and display of results. This chapter explains the firmware design, development environment, signal processing steps, and the estimation algorithm used in the system.

5.2 Development Environment

The firmware is developed using the Arduino IDE (or PlatformIO with Visual Studio Code) for simplicity and ease of implementation. The system is built around a microcontroller (such as ESP32/Arduino) interfaced with the MAX30102 optical sensor.

The key libraries used are:

- **MAX30102 Library:** Used for interfacing with the optical sensor and acquiring PPG signals.
- **Wire Library**: Handles I2C communication between the sensor and microcontroller.
- **Arduino Core Functions**: Basic functions such as pinMode, digitalWrite, analogRead, and delay.
- **Signal Processing Code:** Custom-written functions for filtering, peak detection, and DC removal.

These tools provide an efficient platform for real-time data acquisition and processing.

5.3 Haemoglobin Estimation Algorithm

The firmware implements a signal-processing-based estimation algorithm as the core logic. The system acquires red and infrared PPG signals and processes them to estimate haemoglobin concentration.

The algorithm follows these steps:

- Acquire raw red and infrared signals from the MAX30102 sensor.
- Separate AC (pulsatile) and DC (baseline) components
- Apply filtering techniques to remove noise and motion artifacts.
- Calculate the ratio of ratios (R) using red and infrared signals.
- Use a regression model to estimate haemoglobin (Hb) value

The haemoglobin value is then classified into three levels:

- Low (Anaemia): Hb below normal range
- Normal: Hb within healthy range
- High: Hb above normal range

Haemoglobin Estimation Formula

$Hb = aR + b$

Where:
Hb = Haemoglobin concentration (g/dL)
R = Ratio of red and infrared signal components
a, b = Calibration constants obtained experimentally

Haemoglobin classification levels are mentioned in the table 5.1

Table 5.1: Haemoglobin classification levels

Parameter	**Normal Range**	**Warning Level**	**Critical Level**
Haemoglobin level	12 – 16 g/dL	10 – 12 g/dL	< 10 g/dL

The system uses threshold-based classification to indicate the health condition. If the Hb value falls below the threshold, an anaemia alert is generated. This simple logic ensures quick and easy interpretation of results.

Chapter 6

Working Principle

6.1 Introduction

This chapter explains the working of the non-invasive anaemia detection system from the moment it is powered on to continuous monitoring and haemoglobin estimation. The system is based on optical sensing principles and signal processing techniques. It captures physiological signals from the fingertip and converts them into meaningful haemoglobin (Hb) values.

6.2 Optical Basis of Haemoglobin Detection

The system works based on the Beer–Lambert Law, which states that light absorption depends on the concentration of the absorbing substance. In this project, haemoglobin present in blood absorbs light at specific wavelengths.

6.2.1 Light Absorption by Haemoglobin

Haemoglobin absorbs light differently at red (660 nm) and infrared (940 nm) wavelengths. When light passes through the fingertip, part of it is absorbed by blood and the remaining light is detected by the sensor. The variation in absorption helps in estimating haemoglobin concentration. Higher Hb levels result in greater absorption of light.

6.2.2 Photoplethysmogram (PPG) Signal

The **MAX30102** sensor captures PPG signals, which represent the variation of blood volume with each heartbeat. The signal consists of:

- AC Component: Represents pulsatile blood flow due to heartbeat
- DC Component: Represents constant absorption by tissues and non-pulsatile blood These components are used to extract useful features for haemoglobin estimation.

6.2.3 Ratio Calculation

The ratio of red and infrared signals is calculated to remove external factors like skin thickness and sensor placement.
This ratio is given by:

$$R = \frac{\left(\frac{AC_{red}}{DC_{red}}\right)}{\left(\frac{AC_{IR}}{DC_{IR}}\right)}$$

This value is used as an input to the haemoglobin estimation.

6.3 System Operation Sequence

The system follows a continuous operation cycle as described below:

- **Power-Up and Initialization:**

The microcontroller initializes the MAX30102 sensor and sets up communication.

- **Sensor Placement:**

The user places a fingertip on the sensor for measurement.

- **Signal Acquisition**:

Red and infrared light signals are transmitted and received continuously.

- **Signal Processing**:

Filtering, DC removal, and peak detection are applied to reduce noise.

- **Feature Extraction**:

AC and DC components are calculated and the ratio is computed.

- **Haemoglobin Estimation**:

The calculated ratio is used in a regression model to estimate Hb value.

- **Display Output**:

The haemoglobin level is displayed along with status indication.

- **Loop Repeat**:

The process repeats continuously for real-time monitoring.

6.4 Haemoglobin Estimation Logic

The system provides simple visual alerts based on haemoglobin levels:
Anaemia (Critical State):Indicator ON (e.g., red LED) → Immediate attention required
Warning State:Indicator ON (e.g., yellow LED) → Mild deficiency
Normal State:Indicator OFF or green indication → No action required

6.5 Summary

The working principle of the system combines optical sensing, signal processing, and regression modeling to estimate haemoglobin levels non-invasively. The system operates continuously and provides real-time feedback, making it suitable for portable and point-of care applications.

Chapter 7

Results and Discussions

The non-invasive anaemia detection system demonstrated reliable performance under both controlled and real-time conditions, achieving consistent haemoglobin estimation with satisfactory accuracy. The integrated optical sensing system performed effectively, with the MAX30102 sensor capturing stable red and infrared signals from the fingertip. The use of averaging and filtering techniques ensured noise reduction, enabling accurate computation of the ratio (R) used for haemoglobin estimation. This coordinated operation allowed reliable prediction of haemoglobin levels based on optical signal patterns.

Real-time testing showed stable system behaviour with proper finger placement, and the OLED display provided immediate and clear output of haemoglobin values along with anaemia classification. The system maintained good consistency across repeated trials, demon-strating its suitability for preliminary health screening applications. The calibration equation derived from regression analysis ensured that the estimated values closely followed expected trends.

During continuous operation, certain challenges were observed that required consideration. Variations in ambient light, finger pressure, and skin characteristics affected sensor readings, leading to slight fluctuations in the ratio values. Additionally, the introduction of a protective enclosure caused a shift in readings, which was addressed using offset correction techniques. These observations highlight the importance of proper calibration and controlled measurement conditions.

The modular design of the system allowed improvements during testing. Signal averaging significantly enhanced stability, while calibration adjustments improved the accuracy of haemoglobin estimation. The system design also allows future integration of advanced algorithms for better prediction and adaptability to varying user conditions.

Comparative analysis shows that the proposed system offers clear advantages over conventional invasive methods by eliminating the need for blood sampling. It provides a quick, pain-less, and cost-effective solution for anaemia screening. Although it does not replace laboratory testing, it serves as an efficient tool for early detection and monitoring.

Operational testing provided valuable insights for further enhancement. Optimization of threshold values and calibration models can improve accuracy and reduce errors. Expanding the dataset used for calibration can further strengthen the reliability of the system across different populations.

The system also demonstrated adaptability under different usage conditions. It maintained consistent performance across multiple users, indicating its potential applicability for general-purpose screening. However, variations due to individual differences suggest the need for personalized calibration in future versions.

Graphical analysis as shown in figure 7.1, of the ratio versus haemoglobin values confirmed a strong inverse linear relationship, validating the use of regression-based calibration. The high R^2 value indicated good model accuracy and reliability in predicting haemoglobin levels from optical data.

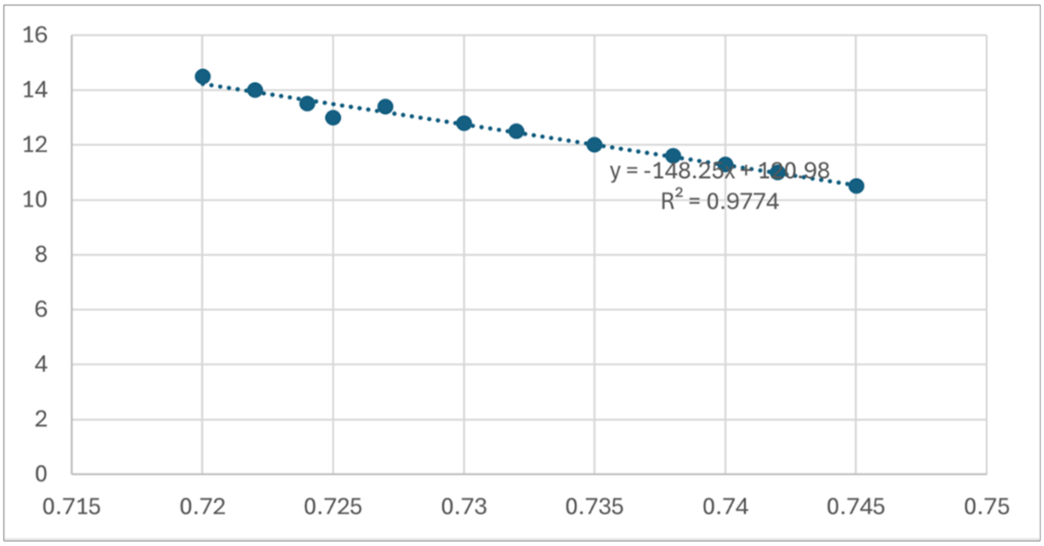


Figure 7.2 : Graphical analysis

Overall, the results validate the effectiveness of the proposed system while identifying areas for improvement. The system's portability, ease of use, and non-invasive nature make it a promising solution for preliminary anaemia detection. With further refinement and advanced calibration techniques, the system can achieve higher accuracy and broader applicability in real-world healthcare scenarios.

The functional Prototype is shown in the figure mentioned as 7.2

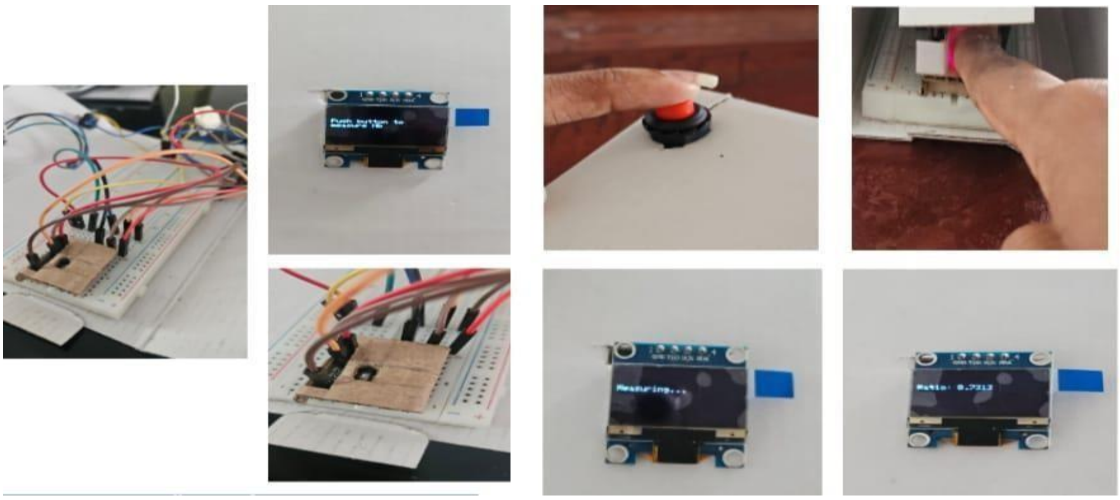


Figure 7.1:Functional Prototype

Chapter 8

Future Scope

The proposed non-invasive anaemia detection system demonstrates the feasibility of estimating haemoglobin levels using optical sensing and basic signal processing. However, there is significant scope for further improvement and enhancement to make the system more accurate, reliable, and suitable for real world applications.

One of the major areas of future work is improving the accuracy of haemoglobin estimation. This can be achieved by collecting a larger dataset from diverse users and applying advanced machine learning algorithms instead of a simple regres-sion model. Techniques such as neural networks or support vector machines can help in capturing complex relationships between optical signals and haemoglo-bin concentration, thereby improving prediction performance.

Another important enhancement is the integration of additional sensors to im-prove reliability. Factors such as skin tone, finger thickness, and ambient light can affect optical readings. By incorporating sensors for temperature, ambient light compensation, or multi-wavelength LEDs, the system can reduce errors and provide more consistent results under different conditions.

The system can also be extended to include wireless connectivity features such as Bluetooth or Wi-Fi. This would allow real-time data transmission to smartphones or cloud platforms for remote monitoring and data storage. A mo-bile application can be developed to display historical trends, provide health in-sights, and generate alerts for users and healthcare providers.

Miniaturization and wearable implementation is another promising direction. The current prototype can be redesigned into a compact wearable device such as a finger clip or smartwatch-like system. This would enable continuous monitor-ing of haemoglobin levels and improve user convenience, especially for patients requiring regular screening.

Chapter 9

Conclusion

This project presented the design and development of a low-cost, non-invasive anaemia detection system using optical sensing principles. By applying the Beer–Lambert Law and using red and infrared light, the system estimates haemoglobin concentration through a fingertip without the need for blood sampling. The MAX30102 sensor was used to acquire PPG signals, and basic signal processing techniques such as filtering, DC removal, and peak detection were implemented to improve signal quality.

A regression-based model was developed to estimate haemoglobin levels from the extracted features. The system provides results in g/dL along with simple indicators to classify the condition as normal, warning, or anaemic. This makes the device easy to use and suitable for quick screening in point-of-care or home environments.

Although the system is not intended for clinical diagnosis, it successfully demonstrates the feasibility of non-invasive haemoglobin estimation using affordable hardware and embedded processing. The project highlights the potential of optical sensing technology in improving accessibility to basic healthcare monitoring and serves as a strong foundation for future enhancements and real-world applications.

Appendix A

MAIN CODE

```
include <Arduino.h>

#include <Wire.h>

#include <MAX30105.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128

#define SCREEN_HEIGHT 64

#define OLED_RESET -1

Adafruit_SSD1306 display(SCREEN_WIDTH,SCREEN_HEIGHT, &Wire, OLED_RESET);

 MAX30105 sensor;

// Button pin

#define BUTTON_PIN 15

 // Function to show message void

showMessage(String msg) {

display.clearDisplay();

display.setTextSize(1);

dis
```

```
play.setTextColor(WHI
TE); display.setCur-
sor(0, 20); dis-
play.println(msg); dis-
play.display();
}
{
 Serial.begin(115200);
Wire.begin();
 pinMode(BUTTON_PIN, INPUT_PULLUP);
//OLED
if(!display.begin(SSD1306_SWITCH-
CAPVCC, 0x3C)) { Serial.println("OLED not
found"); while (1);
 }

// MAX30102 init if
(!sensor.begin(Wire,
I2C_SPEED_FAST)) {
```

Se-

rial.println("MAX30102

not found"); showMes-

sage("Sensor Error");

while (1); }

sensor.setup();

showMessage("Push

button to\nmeasure

Hb");

} void loop() { // Wait

for button press if (digi-

talRead(BUTTON_PIN)

== LOW) {

delay(300); // debounce

showMessage("Measur-

ing..."); float sumRed

= 0; float sumIR = 0;

int count = 0; unsigned

long startTime = mil-

lis();

```
// Collect data for 7 sec-
onds  while (millis() -
startTime < 7000) {
long red = sen-
sor.getRed();  long ir =
sensor.getIR();
 // Finger detection  if (ir
> 50000) {  sumRed +=
red;  sumIR += ir;
count++;
sensor.setup();
showMessage("Push
button to\nmeasure
Hb");
}  void loop() { // Wait
for button press  if (digi-
talRead(BUTTON_PIN)
== LOW) {
```

```
delay(300);  // debounce
showMessage("Measur-
ing...");   float sumRed
= 0;  float sumIR = 0;
int count = 0; unsigned
long startTime = mil-
lis();


// Collect data for 7 sec-
onds  while (millis() -
startTime < 7000) {
long red = sen-
sor.getRed();  long ir =
sensor.getIR();
 // Finger detection  if (ir
> 50000) {  sumRed +=
red;  sumIR += ir;
count++;
 }   delay(50);
 }
 // Avoid division error
if (count == 0) {
```

```
showMessage("Place
finger\nproperly"); de-
lay(2000); showMes-
sage("Push button
to\nmeasure Hb");



return;
}
 float avgRed = sumRed
/ count; float avgIR =
sumIR / count;

 float R = (avgRed /
avgIR)+0.01;
// Hb calculation using
calibration equation
float Hb = -85 *R + 75;
 //float Hb = -148.25 *
R + 120.98;
String status;
```

```
// Female thresholds  if
(Hb > 12.0) {  status =
"Normal";
}   else if (Hb > 10.0) {
status = "Mild Anemia";
}  else if (Hb > 7.0) {

status = "Moderate";

}
else {  status = "Se-
vere";
}
Serial.print("Avg Red:
");
Serial.println(avgRed);
Serial.print("Avg IR: ");
Serial.println(avgIR);
Serial.print("Ratio: ");
Serial.println(R);
Serial.print("Hb: ");
Serial.print("Status: ");
```

```
Serial.println(status);
Serial.println(Hb);


// Display ratio  dis-
play.clearDisplay();


display.setTextSize(1);
display.setCursor(0, 20);
display.print("Hb: ");



display.print(Hb, 2); dis-
play.println(" g/dL");
display.print("R: ");
display.println(R, 4);
display.print("Status: ");
diSplay.println(status);
display.display();  de-
lay(10000);  showMes-
sage("Push button
to\nmeasure Hb");
}
```

PLATFORMIOCODE

[env:esp32dev] platform

= espressif32 board =

esp32dev framework =

arduino lib_deps = ada-

fruit/Adafruit GFX Li-

brary@^1.12.5

sparkfun/Spark-

FuMAX3010x Pulse

and Proximity Sensor

Library@^1.1.2 ada-

fruit/Adafruit SSD1306

upload_speed=115200

monitor_speed=11520

Appendix B

Expense Analysis

Complete expense analysis of the project is given in Table A.1

Table A.1: Expense Analysis

No.	Component	Qty	Unit Price (INR)	Total (INR)
1	ESP32 DevKit V1	1	415	415
2	MAX30102	1	245	245
3	OLED Display	1	330	330
4	Breadboard (830-point)	1	80	80
5	Jumper Wire Set	1	50	50
			Total	1120

Appendix C

Project Timeline (Road Map)

Table B.1: Project Timeline

Activity	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**	**W9**
Literature Survey	•	•							
Data Collection		•	•						
System Design			•	•					
Programming				•	•				
Implementation					•	•			
Testing & Calibration						•	•		
Report Writing							•	•	•

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